Development effects of electrification in Brazil

Identification method and replication of main results of *Lipscomb et al (2013)*

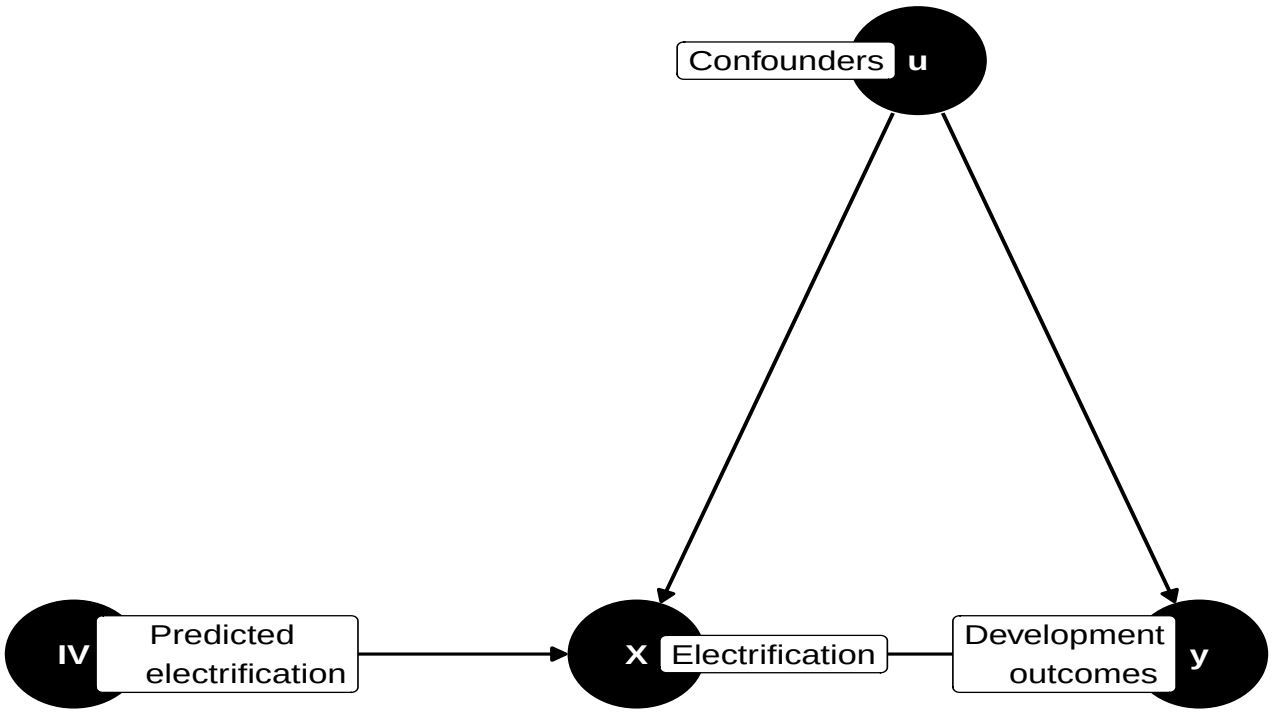
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Throwback to the paper's research design

Research question: What's the effect of electrification on local economic development in Brazil?

- The paper relies on a natural experiment design by using an Instrumental Variable on the prediction of electricity availability.



Identification strategy: fixed-effects IV (2SLS)

$$\textbf{First stage:} \quad E_{c,t-1} = \alpha_c^1 + \gamma_t^1 + \theta\, Z_{c,t-1} + \eta_{ct}$$

$$\textbf{Second stage:} \quad Y_{ct} = \alpha_c^2 + \gamma_t^2 + \beta\, \hat{E}_{c,t-1} + \varepsilon_{ct}$$

County c, decade t, 1960-2000. Electrification is lagged one decade.

Causal identification if:

Relevance of the instrument: met, first-stage $F \approx 25 > 10$.

Exogeneity of the instrument: no relationship between predicted electricity and confounders.

Exclusion restriction: geography affects local development only via electrification.

Monotonicity: not discussed but very plausible to hold.

E: proportion of grid points in a county that are electrified in period t
Z: proportion of grid points in a county predicted to be electrified by the model
$\hat{E}$: instrumented electricity provision, predicted from the first stage

Main findings and replication

- **Housing values**: +10 pp electrification → **+881 reais**. It's a change of +6.8% of the mean value (Table 6).
- **HDI**: moving from 0 to full electrification → **+0.11**. Would move the median county towards the 69th percentile county (Table 7).
  TABLE

Are the results convincing?

Convincing

- IV simulates electricity expansion leaving out local demand.
- Results are robust after controls for the geographic factors used to construct the IV.

Not convincing

Omitted variables: grid may come with package of infrastructure. Only partially addressed as roads are only proxied.

Mechanisms: migration data only for 1990-2000, while most dams were built earlier $\rightarrow$ possible missing variable between X and Y

Dam effects: changes in local hydrology might affect local economic development (flood control)

Measurement: all dams are considered the same size and with a fixed 50 km reach. HDI is the only welfare measure.

Appendix: Research design: ideal benchmark vs. the paper

1. RCT: randomized grid rollout

Among areas where grid expansion is feasible, a lottery decides which get the grid now and which wait ten years.

2. Staggered diff-in-diff

Valid if construction timing depended on administrative factors unrelated to local growth (parallel trends). Here assignment is not random so is not plausible.

2b. Instrumented diff-in-diff (TBD)

Combine staggered timing with an instrument for when areas are electrified, to deal with endogenous placement. Possible avenue to research.

Appendix: Table 6

Appendix: Table 7

Appendix: Table 10